



**BAULKHAM HILLS HIGH SCHOOL**

**2015**

**YEAR 11**

**TASK 2 JULY ASSESSMENT TASK**

# Mathematics

## General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Black pen is preferred
- Board-approved calculators may be used
- All relevant mathematical reasoning and/or calculations must be shown

**Total marks – 45**

(Pages 2-5)

Attempt Questions 1-4

**Total Marks – 45**

**Attempt Questions 1 – 4**

Answer each question on the appropriate page of your answer booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

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**Question 1 (12 marks)** Start on the appropriate page of your answer booklet

- a) A line makes an angle of  $60^\circ$  with the positive  $x$ -axis. What is the equation of the line if it passes through the point  $(-1,4)$ ? 2

b) Evaluate:

(i)  $\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3}$  2

(ii)  $\lim_{x \rightarrow \infty} \frac{x^2 + 2}{2x^2 - 3x + 1}$  2

c) Differentiate

(i)  $\frac{1}{4x + 3}$  2

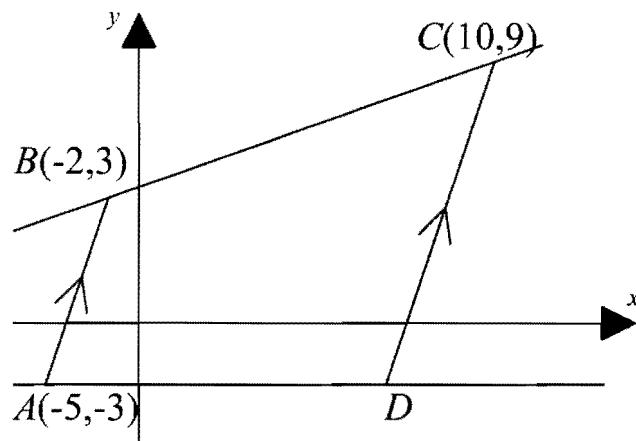
(ii)  $\frac{2x}{x^2 + 3}$  2

(iii)  $\sqrt{2x^4 - 7x^3 + 1}$  2

**END OF QUESTION 1**

**Question 2 (12 marks)** Start on the appropriate page of your answer booklet

- a) On a number plane the points  $A(-5,-3)$ ,  $B(-2,3)$ ,  $C(10,9)$  and  $D$  form a trapezium in which  $AD$  is parallel to the  $x$  axis and  $AB \parallel CD$ .



- |       |                                                                                                           |   |
|-------|-----------------------------------------------------------------------------------------------------------|---|
| (i)   | Show that the equation of the line $DC$ is $2x - y - 11 = 0$                                              | 2 |
| (ii)  | Find the coordinates of $D$ .                                                                             | 1 |
| (iii) | Find the perpendicular distance from $B$ to $DC$                                                          | 2 |
| (iv)  | Find the coordinates of the midpoint of $BD$ .                                                            | 1 |
| (v)   | The point $E$ lies on the line $BA$ produced such that $BCDE$ is a rhombus. Find the coordinates of $E$ . | 1 |
- b) Joel is trying to find the equation of a line perpendicular to  $x - 2y + 13 = 0$  which passes through the point of intersection of the lines  $3x + 2y - 1 = 0$  and  $7x - 11y + 3 = 0$ .

As part of his working Joel has correctly set up the following equation:

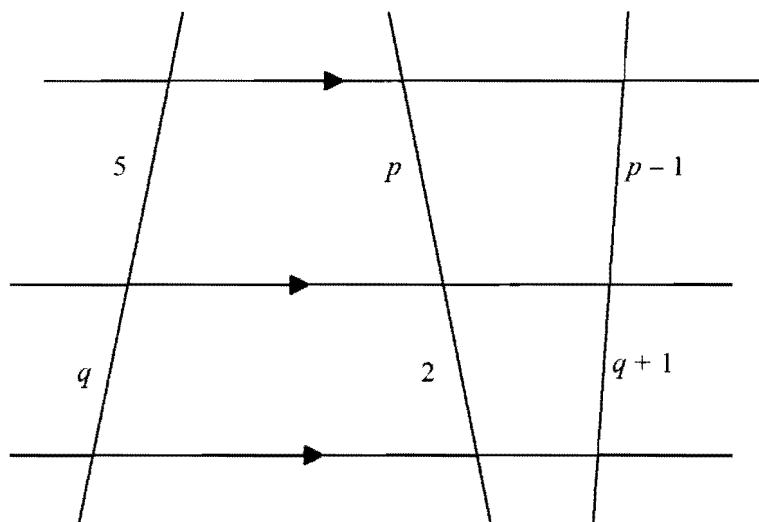
$$3x + 2y - 1 + k(7x - 11y + 3) = 0$$

- |       |                                                                |   |
|-------|----------------------------------------------------------------|---|
| (i)   | Show that the gradient of the line is $\frac{3 + 7k}{11k - 2}$ | 2 |
| (ii)  | Find the value of $k$                                          | 2 |
| (iii) | Hence determine the required equation of the line.             | 1 |

**END OF QUESTION 2**

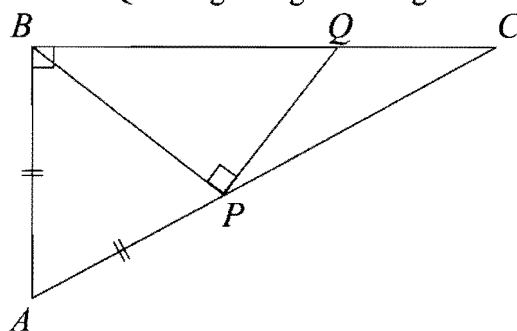
**Question 3 (12 marks)** Start on the appropriate page of your answer booklet

- a) The diagram below comprises three parallel lines.



- i) Explain why  $\frac{5}{q} = \frac{p}{2}$  1
  - ii) Find the exact value of  $p$  and  $q$  2
- b) Find the equation of the tangent to the curve  $y = 3x^3 - 4x^2 + x - 1$  at the point where  $x = 1$ . 3

- c) Triangles ABC and BPQ are right angled triangles and  $AB=AP$



Copy the diagram into your answer booklet

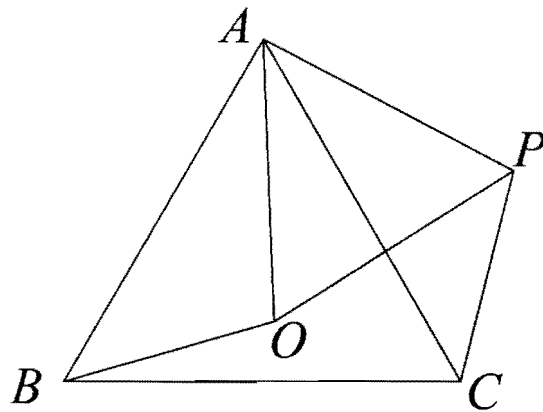
- (i) Prove  $\angle CBP = \angle CPQ$  3
- (ii) Hence prove that  $PC^2 = BC \times QC$  3

**END OF QUESTION 3**

**Question 4 (9 marks)** Start on the appropriate page of your answer booklet

- a) The parabola  $y = ax^2 + bx + c$  passes through the point  $(0, -2)$ . At the point  $(2, -3)$  on this curve, the equation of the tangent is  $2x - y - 7 = 0$ .  
Find the values of  $a$ ,  $b$  and  $c$ .

- b) In the figure triangles  $ACB$  and  $APC$  are equilateral.



*NOT TO SCALE*

Copy this diagram into your answer booklet.

- |       |                                            |   |
|-------|--------------------------------------------|---|
| (i)   | Prove that $\angle BAO = \angle PAC$       | 2 |
| (ii)  | Prove $\triangle AOB \equiv \triangle APC$ | 3 |
| (iii) | Hence, prove $OB = CP$ .                   | 1 |

**END OF EXAMINATION**

YR 11 JUNE 2015 2 UNIT

Q1

a)  $m = \tan 60^\circ$   
 $= \sqrt{3}$  ✓

$$y - y_1 = m(x - x_1)$$

$$y - 4 = \sqrt{3}(x + 1)$$

$$y = \sqrt{3}x + \sqrt{3} + 4$$
 ✓  
 (OR  $\sqrt{3}x - y + \sqrt{3} + 4 = 0$ )

- (2) correct solution  
 (1) finding gradient  
 or using incorrect gradient  
 to correctly find line

b) i)  $\lim_{x \rightarrow 3} \frac{x^2 - 8x + 15}{x - 3}$   
 $= \lim_{x \rightarrow 3} \frac{(x - 5)(x - 3)}{x - 3}$   
 $= \lim_{x \rightarrow 3} (x - 5)$  ✓  
 $= 3 - 5$   
 $= -2$  ✓

- (2) correct solution  
 (1) incorrect factorisation  
 (1) and subsequent cancelling

ii)  $\lim_{x \rightarrow \infty} \frac{x^2 + 2}{2x^2 - 3x + 1}$   
 $= \lim_{x \rightarrow \infty} \frac{\frac{x^2}{x^2} + \frac{2}{x^2}}{\frac{2x^2}{x^2} - \frac{3x}{x^2} + \frac{1}{x^2}}$  ✓  
 $= \lim_{x \rightarrow \infty} \frac{1 + \frac{2}{x^2}}{2 - \frac{3}{x} + \frac{1}{x^2}}$   
 $= \frac{1 + 0}{2 - 0 + 0}$   
 $= \frac{1}{2}$  ✓

- (2) correct solution  
 with/without working  
 (1) attempts to use  $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$   
 or (1) divides throughout by  $x^2$

$$1c \ i) \ \frac{d}{dx} (4x+3)^{-1} = -1 (4x+3)^{-2} \cdot 4 \checkmark$$

$$= \frac{-4}{(4x+3)^2} \checkmark$$

(2) correct solution

(1) differentiation without multiplying by 4.

$$ii) \ \frac{d}{dx} \left( \frac{2x}{x^2+3} \right) = \frac{(x^2+3) \cdot 2 - 2x \cdot 2x}{(x^2+3)^2} \checkmark$$

$$= \frac{2x^2+6-4x^2}{(x^2+3)^2}$$

(2) correct solution

(1) attempts to use quotient rule

$$= \frac{6-2x^2}{(x^2+3)^2} \text{ or } \frac{2(3-x^2)}{(x^2+3)^2} \checkmark$$

$$(iii) \ \frac{d}{dx} (2x^4-7x^3+1)^{\frac{1}{2}} = \frac{1}{2} (2x^4-7x^3+1)^{-\frac{1}{2}} \cdot (8x^3-21x^2) \checkmark$$

$$= \frac{8x^3-21x^2}{2\sqrt{2x^4-7x^3+1}} \text{ or } \frac{x^2(8x-21)}{2\sqrt{2x^4-7x^3+1}} \checkmark$$

(2) correct soln

(1) differentiates without using chain rule

$$2 \ a) i) \ m_{AB} = m_{AB}$$

$$= \frac{3-(-3)}{-2-(-5)}$$

$$= \frac{6}{3}$$

$$= 2 \checkmark$$

(2) correct solution

(1) correct gradient or finds equation of line using incorrect gradient

$$\text{Using } y-y_1 = m(x-x_1)$$

$$y-9 = 2(x-10)$$

$$y = 2x-20+9$$

$$y = 2x-11 \text{ or } 2x-y-11=0 \checkmark$$

$$ii) \ \text{when } y=-3$$

$$2x + 3 - 11 = 0$$

$$2x = 8$$

$$x = 4$$

$$\therefore D \text{ is } (4, -3) \checkmark$$

(1) correct solution

2a (iii) Perp Dist:  $\left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$

$$= \left| \frac{2x - 2 - 3 - 11}{\sqrt{2^2 + 1^2}} \right| \quad \checkmark$$

$$= \left| \frac{-18}{\sqrt{5}} \right|$$

$$= \frac{18\sqrt{5}}{5} \quad \checkmark$$

(2) correct solution  
(1) substitutes into perpendicular distance formula.

(iv) Midpoint =  $\left( \frac{-2+4}{2}, \frac{3+(-3)}{2} \right)$   
 $= (1, 0) \quad \checkmark$

(1) correct solution

(v)  $\begin{cases} 1 = \frac{x+10}{2} \\ 0 = \frac{y+9}{2} \end{cases}$

(1) correct solution

$$2 = x+10$$

$$x = -8$$

$$y+9=0$$

$$\therefore E \text{ is } (-8, -9) \quad y = -9 \quad \checkmark$$

2b (i)  $3x+2y-1 + 7kx - 11ky + 3k = 0$   
 $(7k+3)x + 3k-1 = (11k-2)y$   
 $\therefore m = \frac{7k+3}{11k-2} \quad \checkmark$

(2) correct solution  
(1) expresses in general form or  $y=mx+b$

(ii) grad  $\perp = -2 \therefore -2 = \frac{7k+3}{11k-2} \quad \checkmark$

$$-2(11k-2) = 7k+3$$

$$-22k+4 = 7k+3$$

$$-29k = -1$$

$$k = \frac{1}{29} \quad \checkmark$$

$4y = 11x+3$   
 $y = \frac{11}{4}x + \frac{3}{4}$   
 $m_2 = -2$  (2) correct solution  
 (1) equating gradient with -2

(iii)  $3x+2y-1 + \frac{1}{29}(7x-11y+3) = 0$   
 $87x+58y-29+7x-11y+3=0$   
 $94x+47y-26=0 \quad \checkmark$

(1) correct solution



3a) (i)  $\frac{5}{q} = \frac{p}{2}$  (ratio of intercepts on parallel lines) ① correct reason

(ii)  $\frac{5}{q} = \frac{p}{2}$   
 $10 = pq$

$\frac{p}{2} = \frac{p-1}{q+1}$   
 $pq + p = 2p - 2$  ②  
 $10 + p = 2p - 2$   
 $12 = p$   
 $\therefore q = \frac{10}{p}$   
 $= \frac{10}{12}$   
 $q = \frac{5}{6}$

② correct solution  
 ① uses ratio of intercept theorem to set up simultaneous equations and attempts to solve

b)  $\frac{dy}{dx} = 9x^2 - 8x + 1$

when  $x=1$ ,  $\frac{dy}{dx} = 9 - 8 + 1$   
 $= 2$   
 $f(1) = 3 - 4 + 1 = 0$

$y + 1 = 2(x - 1)$   
 $y + 1 = 2x - 2$   
 $y = 2x - 3$  or  $2x - y - 3 = 0$  ✓

③ correct solution  
 ② either gradient or y-value or equation contains error  
 ① finds gradient or y-value or equation.

c) i) let  $\angle PBQ = x$   
 $\angle PBA = 90^\circ - x$  (adjacent complementary  $\angle$ 's) ✓  
 $\angle BPA = 90^\circ - x$  (equal  $\angle$ 's opposite equal sides in  $\triangle PAB$ )  
 $AB = AP$  given  
 $\angle APB + \angle BPQ + \angle QPC = 180^\circ$  (angle sum of straight line) ✓  
 $90^\circ - x + 90^\circ + \angle QPC = 180^\circ$   
 $\angle QPC = 180^\circ - 90^\circ + x = 90^\circ + x$

③ correct solution  
 ② correct with at least 1 reason  
 ① correct with no reasons or expression for  $\angle BPA$  with reason  
 $\angle PBQ$  as req'd. ✓

C (iii) In  $\triangle BCP$  and  $\triangle PCQ$

$LC$  is common

$$\angle CBP = \angle PCQ \quad (\text{proven in (ii)})$$

$\therefore \triangle BCP \parallel \triangle PCQ$  (two pairs of matching angles equal) ✓

$$\frac{BC}{PC} = \frac{PC}{QC} \quad (\text{matching sides of similar } \triangle BCP \text{ \& } \triangle PCQ \text{ in same ratio}) \quad \checkmark$$

$$\therefore PC^2 = BC \cdot QC$$

- ③ correct solution
- ② correct with reason
- ① correct with no reasons or proves similar without reasons

Q4

a)

$$y = ax^2 + bx + c$$

since curve passes through  $(0, -2)$

$$-2 = 0 + 0 + c$$

$$\therefore y = ax^2 + bx - 2$$

since  $(2, -1)$  lies on curve

$$-1 = 4a + 2b - 2$$

$$-1 = 4a + 2b$$

$$4a + 2b = -1$$

$$\frac{dy}{dx} = 2ax + b$$

When  $x = 2$ ,  $\frac{dy}{dx} = 2$  gradient of tangent

$$\therefore 4a + b = 2$$

- ③ correct
- ② forms simultaneous equations and attempts to solve (with c)
- ① finds c or  $4a + 2b = -1$  or  $4a + b = 2$

①-②

$$b = -3$$

sub in ①  $4a - 3 = 2$

$$4a = 5$$

$$a = \frac{5}{4}$$

$$\therefore y = \frac{5}{4}x^2 - 3x - 2$$

4 b(i)

Let  $\angle OAC = x$

$\angle BAC = \angle OAP = 60^\circ$  (angles in equilateral  $\Delta$ 's)

$\angle BAO = 60^\circ - x$

$\angle PAC = 60^\circ - x$

$= \angle BAO$

$\therefore \angle BAO = \angle PAC$

Correct solution

(1) Attempt to use  $60^\circ$  angle to find an expression

(ii) In  $\Delta AOB$  and  $\Delta APC$

$AB = AC$  (equal sides of equilateral  $\Delta ABC$ ) ✓

$OA = PA$  (equal sides of equilateral  $\Delta OAP$ ) ✓

$\angle BAO = \angle PAC$  (proven in (i)) ✓

$\therefore \Delta AOB \cong \Delta APC$  (SAS)

(iii)  $OB = CP$  (matching sides in congruent  $\Delta$ 's  $\Delta AOB$   $\Delta APC$ ) ✓